

Turntable

Turntable [object]

Use the object *Turntable* for modeling a rotating platform, which moves the part onto one of several connected material flow objects and/or turns it around.

Image



Description

The *Turntable* has a capacity of one, i.e. only one part can be located on it at any one time. The length of the *MU* must not be longer than the **Length** of the *Turntable* itself.

The *Turntable* moves the part on to its successors within the flow of materials like this:

- The *MU* arrives at the exit of the predecessor. It notifies the turntable that it wants to be rotated.
- The turntable then determines if it accepts the *MU* for rotating it or not.
 - Without a **Pull Control**, the turntable will be notified that a *MU* wants to be rotated, and it accepts it.
 - If you entered a **Pull Control**, the turntable executes it, and when the *MU* is selected, it will be rotated.
- The turntable rotates to the respective predecessor, and the *MU* moves onto the turntable and drives onto it.
- Once one of the conditions below is met, the turntable looks for the target station and starts rotating toward it:
 - If the *MU* has **completely entered** the turntable.
 - If the *MU* has **reached the rotation point** on the table.
 - If the part is located in the **center** of the turntable, i.e., when both ends of the part have the same distance from the ends of the turntable.
 - If you want the turntable to not rotate automatically, but only if the *MU* has reached a user-defined sensor, select **User-defined with Sensor**. In the **Sensor Control**, you have to tell the turntable to call the method **setDestination**.
- For determining the target station, the turntable either uses its default exit strategies or the **Target Control**. Within this *Method* you set the target station with the method *setDestination*.

Note

Do not use an **Exit Control** for determining the target of the *MU*, as an **Exit Control** will be called only when the *MU* is ready to exit the *Turntable*. This point in time is too late though, as the target has to be determined earlier in time.

- Then the turntable rotates to the target and *MU* moves on, once the turntable has reached the final position of the rotation. Only then a new *MU* can move onto the *Turntable*. The *Turntable* has a **Capacity** of 1.

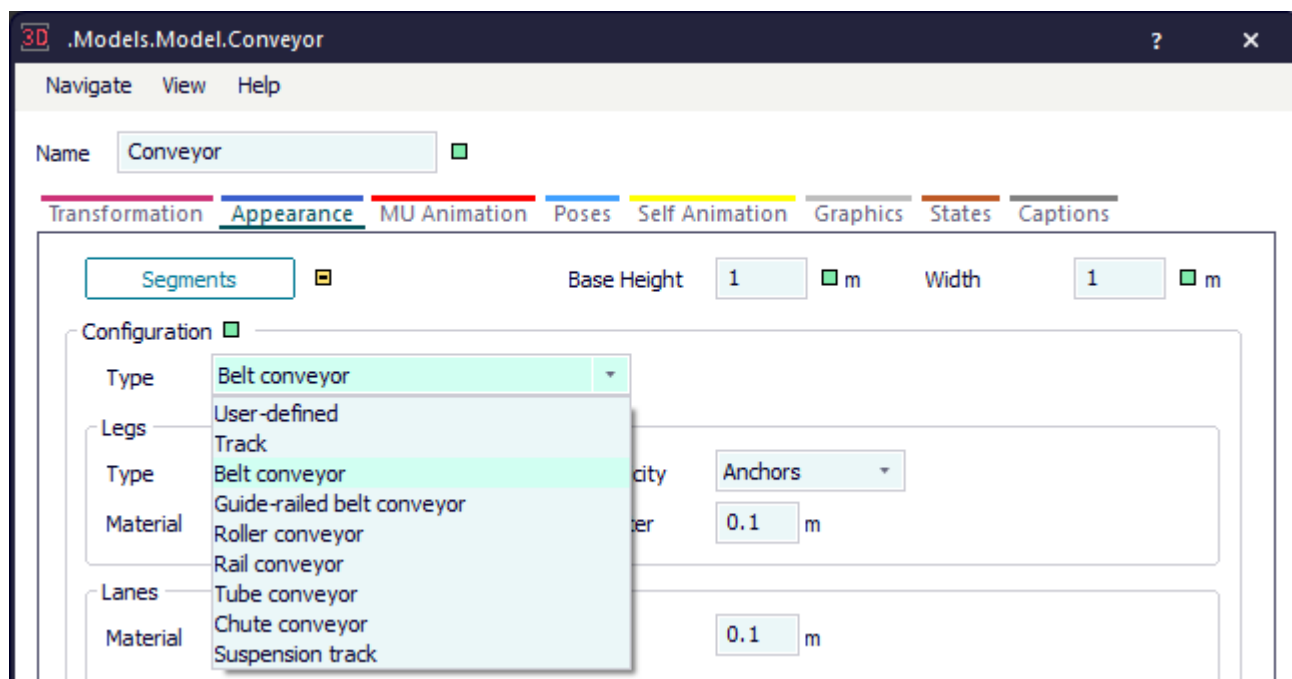
Note

You can query the current rotation angle with the read-only attribute **CurrentAngle [SimTalk]**.

Note

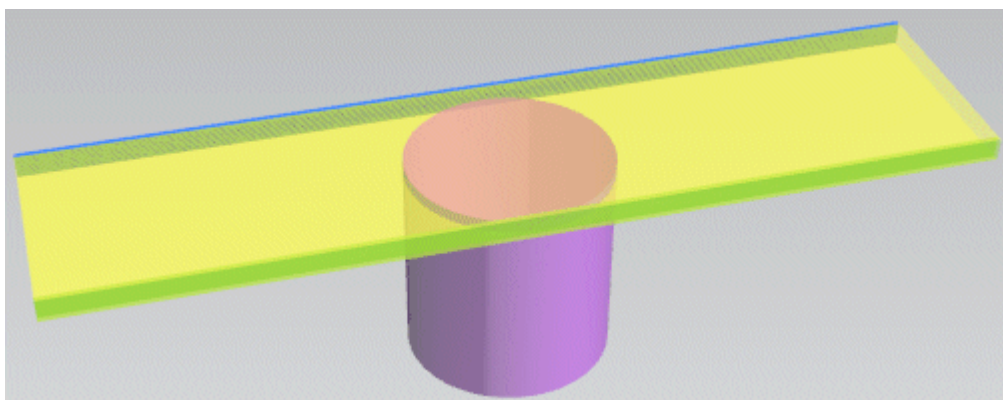
If an object, from which/onto which the *MU* is to be moved, does not exist in the respective angle table, the *Turntable* checks the **Exit Angle Table** for an entering *MU* or the **Entry Angle Table** for a leaving *MU*, if the object exists there and uses that angle. If the object is not contained in the respective table, the *Turntable* uses the angle from the layout in the *Frame*.

You can select different configurations for the *Turntable* on the tab **Appearance of the Length-oriented Objects**.

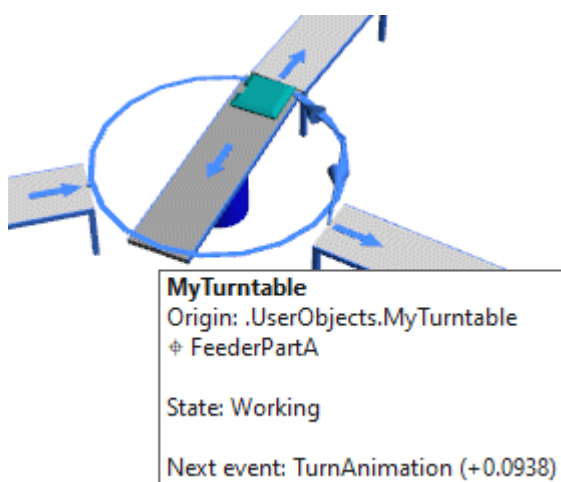


An animatable object with the **Velocity** of 0 does not rotate with the *Turntable* when the *Turntable* rotates. This applies if the insertion position of the animatable object is located on the animation rotation axis of the *Turntable*. This is the case if the insertion point is exactly located on the animation rotation center or is reachable from there along the animation rotation axis without a sideways deviation.

This resembles a real *Turntable* with a fixed base and a rotary table.



To show a **tooltip** with information about the *Turntable*, hover with the mouse over it.



To change the length of the graphic and the anchor points of the *Turntable*, click **Show Manipulators**



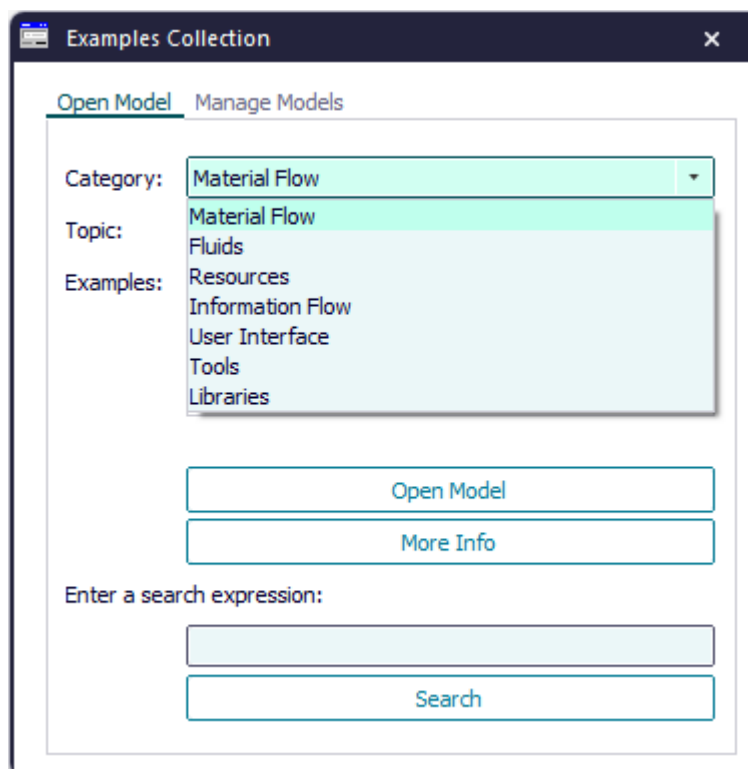
on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model



To add the object *Turntable* to your simulation model, click **Manage Class Library** > **Basic Objects** > **MaterialFlow** > **Turntable** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page** > **Getting Started** > **Example Models** > **Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Move Parts On with the Turntable](#)

Video on YouTube

<https://youtu.be/hOvdrDnvXXo?si=C55gOF4JB5PBIVma&t=532>

Dialog Box of the Turntable

Dialog Box of the Turntable

Double-click the icon of the *Turntable* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the [Tab Attributes](#).

Length [text box] - Turntable

Type the **Length** of the *Turntable* into the text box.

Length:  m

Remarks

After you inserted it, *Plant Simulation* shows its length here. The *Turntable* only rotates parts, which are shorter or as long as the value, which you enter here.

You can change the length of the graphic and the anchor points of the *Turntable* by clicking [Show](#)

[Manipulators](#)  on the **Edit** ribbon tab or by pressing **M** on the keyboard.

SimTalk

Length [SimTalk] - Turntable

See also

[OccupiedLength \[SimTalk\] - Conveyor](#)

Width [text box] - Turntable

Type the **Width** of the *Turntable*.

Width:  m

Remarks

You can change the width of the graphic and the anchor points of the *Turntable* by clicking [Show](#)

[Manipulators](#)  on the **Edit** ribbon tab or by pressing **M** on the keyboard.

SimTalk

Width [SimTalk] - Turntable

Rotation Point [text box]

Type the **Rotation Point** into the text box, i.e., the position of the fulcrum around which the *Turntable* rotates.

Rotation point: m

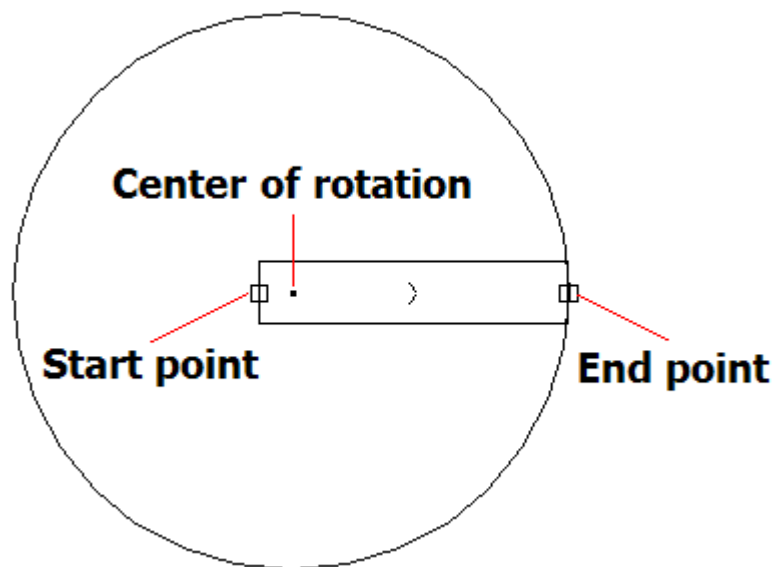
Remarks

You can type in a value between 0 and the length of the *Turntable*. 0 designates the point at which you start inserting the *Turntable*.

The **Center of Rotation** is the fulcrum around which the table rotates.

The **Start Point** is the point which you click, when you start inserting the *Turntable*.

The **End Point** is the point which you click, when you finish inserting the *Turntable*. The circle in the screenshot below shows all points which the turntable can reach with its end point, when it rotates around the fulcrum.



SimTalk

[RotationPoint \[SimTalk\]](#)

Conveyor Speed [text box]

Type the **Conveying Speed** into the text box with which the *Turntable* transports the *MU*, while it is located on the table.

Conveyor Speed: ☐ m/s

SimTalk

[Speed \[SimTalk\] - Turntable](#)

See also

[Velocity \[joint\]](#)

Rotation Time per 90° [text box] - Turntable

Type the **Time** into the text box, which it takes the *Turntable* to rotate by 90 degrees.

Rotation Time per 90°: 

Remarks

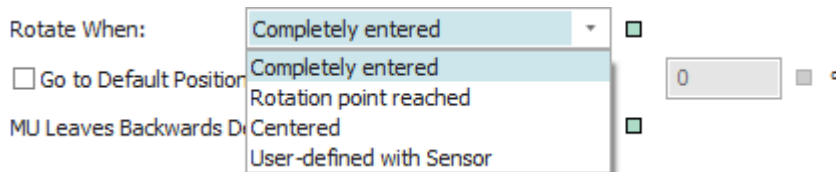
To rotate the *MU* immediately, without using up any time at all, type in a rotation time of 0.

SimTalk

RotationTimePer90Degrees [SimTalk] - Turntable

Rotate When [drop-down list] - Turntable

Select when the *Turntable* rotates toward its target object.



Remarks

You can select one of these settings:

- When the *MU* has **completely entered** the *Turntable*.
- When the *MU* has **reached the rotation point/the fulcrum** on the table.
- When the *MU* is located in the **center** of the *Turntable*, i.e., when both ends of the *MU* have the same distance from the edges of the turntable.
- If you want the *Turntable* to not rotate automatically, but only when the *MU* has reached a user-defined sensor, select **User-defined with Sensor**. In the **Sensor Control**, you have to tell the *Turntable* to call the method [setDestination](#).

SimTalk

[RotateWhen \[SimTalk\]](#)

Go to Default Position [check box] - Turntable

To rotate the *Turntable* back to its **Default Position**, when the *MU* has left the *Turntable* and when no new *MU* is ready to be rotated, select this check box.

☐ Go to Default Position	☐	Default Angle:	0	☐ °	☒ Go to Default Position	☐	Default Angle:	0	☐ °
☐ Only for Empty Blocking List	☐	Blocking Angle:	-1	☐ °	☐ Only for Empty Blocking List	☐	Blocking Angle:	-1	☐ °

Remarks

If you select the check box, the *Turntable* uses the **Default Angle**, when you reset the model.

Then, type in the default angle at which the default position is located.

SimTalk

[GoToDefaultPosition \[SimTalk\] - Turntable](#)

See also

[Default Angle \[text box\] - Turntable](#)

[DefaultAngle \[SimTalk\] - Turntable](#)

Default Angle [text box] - Turntable

Type the **Default Angle** into the text box to which the *Turntable* rotates, when you selected **Go to Default Position**.

☒ Go to Default Position	☐	Default Angle:	0	☐ °
☐ Only for Empty Blocking List	☐	Blocking Angle:	-1	☐ °

Remarks

The default position is located at this angle. If you would like the *Turntable* to rotate to the side on which the start point is located, add 180 degrees to this angle.

SimTalk

DefaultAngle [SimTalk] - Turntable

See also

Go to Default Position [check box] - Turntable

GoToDefaultPosition [SimTalk] - Turntable

MU Leaves Backwards Depends On

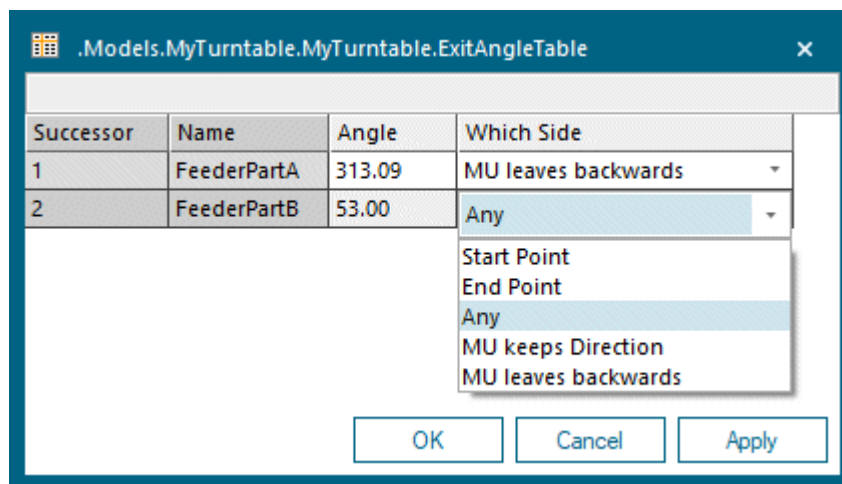
Type the name of the **user-defined attribute** of the *MU*, which triggers its rotation, so that it exits the table driving backward into the text box.

MU Leaves Backwards Depends On: 

Remarks

This **user-defined attribute** can be an attribute of data type *boolean* or a *Method* which returns a value of data type *boolean*.

MU leaves Backwards Depends On will only be evaluated, when you select **Any** as the side in the [Exit Angle Table](#).



SimTalk

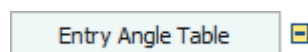
MURotationAttribute [SimTalk]

See also

Exit Angle Table

Entry Angle Table

To open the table of the **Entry Angles** at which the *MUs* move onto the *Turntable*, click this button.



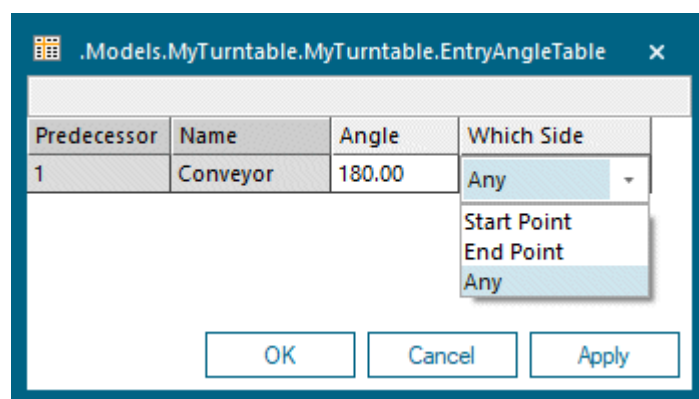
Remarks

You can view or fine-tune these settings:

- The number of the **Predecessor**.
- The name of the **Predecessor**.
- The **Angle** at which the *Connector* from this predecessor connects to the *Turntable*.
- At **Which Side** it turns toward the predecessor.
 - If you select **Start Point**, the *Turntable* turns with the side, at which you started inserting the *Turntable*, toward this predecessor.
 - If you select **End Point**, the *Turntable* turns with the side, at which you finished inserting the *Turntable*, toward this predecessor.
 - If you select **Any**, the *Turntable* computes the smallest angle and turns with this computed side toward this predecessor.

Note

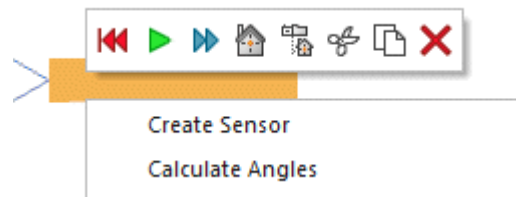
The button **Entry Angle Table** is only active after you connected the *Turntable* with its predecessor.



The figure illustrates the position of the **Entry Angle** depending on the insertion direction of the *Turntable*.

Left to right	Right to left	Top to bottom	Bottom to top

If you want the *Turntable* to compute the angles at which you connected it to its predecessors, click it with the right mouse button and select **Calculate Angles**.



SimTalk

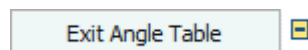
[setEntryAngles \[SimTalk\]](#)

[getEntryAngles \[SimTalk\]](#)

[calculateAngles \[SimTalk\] - Turntable](#)

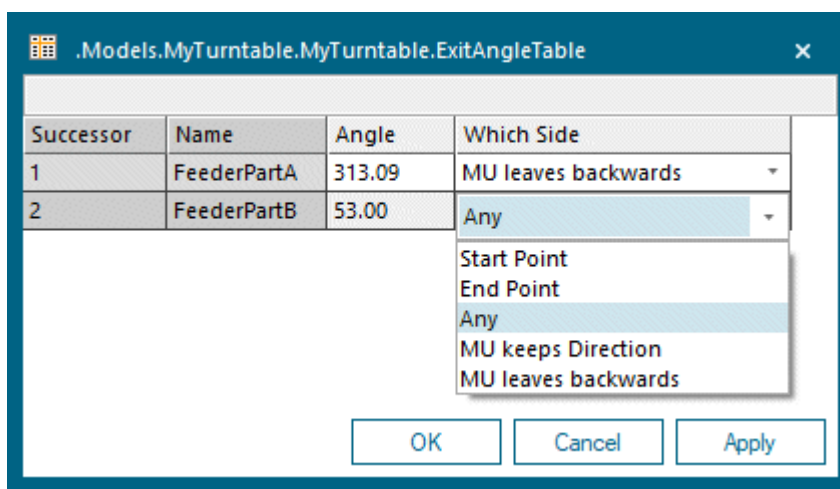
Exit Angle Table

To open the table of the **Exit Angles** at which the *MUs* leave the *Turntable*, click this button.



Remarks

You can view or fine-tune these settings:

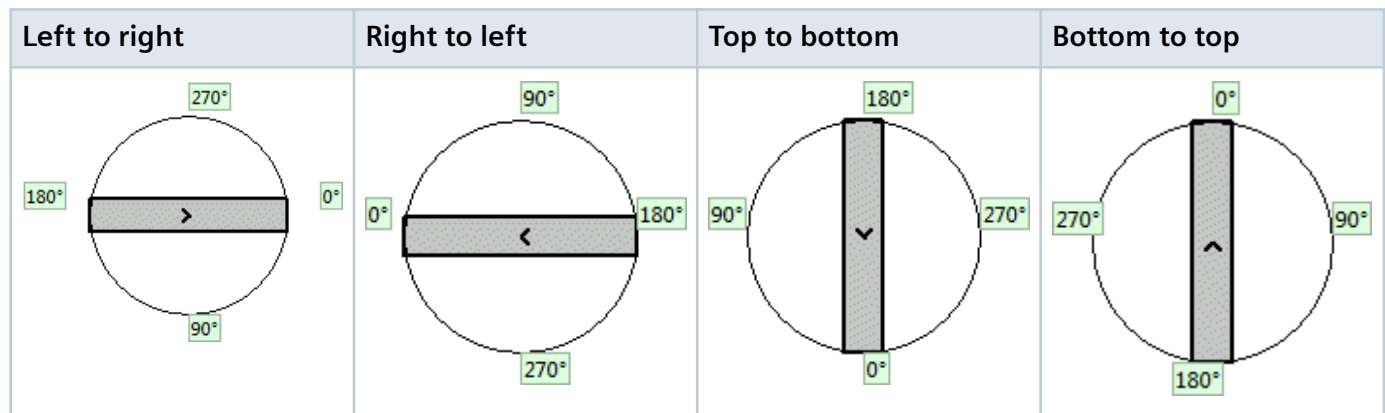


- The number of the **Successor**.
- The name of the **Successor**.
- The **Angle** at which the *Connector* from the *Turntable* connects toward this successor.
- At **Which Side** it turns toward the successor.
 - If you select **Start Point**, the *Turntable* turns with the side, at which you started inserting the *Turntable* toward this successor.
 - If you select **End Point**, the *Turntable* turns with the side, at which you finished inserting the *Turntable* toward this successor.
 - If you select **Any**, the *Turntable* computes the smallest angle and turns with this computed side toward this successor. With this setting the *MU* can leave backward or forward, depending on the side which can be reached with the smallest angle.
 - If you select **MU keeps Direction**, the *Turntable* turns this side of the *Turntable* toward this successor which makes the *MU* keep its direction.
- If you select **MU leaves backward**, the *Turntable* turns this side of the *Turntable* toward this successor so that the *Turntable* turns the *MU* around, and it leaves moving backward. Compare **MU Leaves Backwards Depends On**.

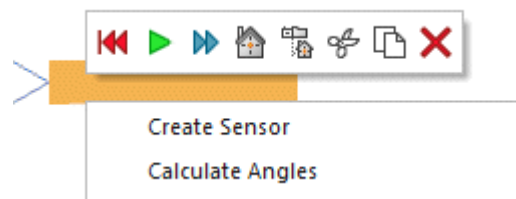
Note

The button **Exit Angle Table** is only active after you connected the *Turntable* with its successor.

The figure illustrates the position of the **Exit Angle** depending on the insertion direction of the *Turntable*.



If you want the *Turntable* to compute the angles at which you connected it to its successors, click it with the right mouse button and select **Calculate Angles**.

**SimTalk**

[setExitAngles \[SimTalk\]](#)

[getExitAngles \[SimTalk\]](#)

[calculateAngles \[SimTalk\] - Turntable](#)

Automatic Stop [check box] - Turntable

To set the **Current Speed** of the *Turntable* to 0 if it does not transport a part, select this check box.



Remarks

Automatically stopping the *Turntable* might, for example, be the case if it is empty or if it is blocked because a *MU* cannot leave it.

If the speed of the *Turntable* is 0, the **Energy State** changes to **Operational**.

SimTalk

AutomaticStop [SimTalk] - Turntable

Tab Times [Turntable]

Define **Times** as described under the **Tab Times**.

Select a distribution from the drop-down list and type the values that this distribution requires into the text box. *Plant Simulation* shows the parameters along the upper border of the tab. You can also select a constant time (**Const**).

Note

The *Turntable* does not provide the **Cycle Time**.

You can set the type of the distribution and a complete set of parameters with the method [setTypeAndAttr \[SimTalk\]](#).

See also

[Recovery Time \[general description\]](#)

[Recovery Time Starts](#)

Tab Failures

Define **failures** as described under the [Tab Failures](#).

Tab Controls

Tab Controls [Turntable]

Provides controls to modify the built-in behavior of the object.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

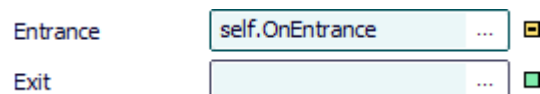
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.

- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

See also

[Entrance Control \[general description\]](#)

[Exit Control \[general description\]](#)

[Pull Control \[general description\]](#)

[Target Control \[Turntable\]](#)

Target Control [Turntable]

Modifies the built-in behavior of the object. The object calls the **Target Control** as soon as the *MU* has completely moved onto the *Turntable* or has reached the center of rotation. As opposed to the **Exit Control**, the *MU* is not ready to exit the object yet.

Remarks

Do not use an **Exit Control** for determining the target of the *MU*, as an **Exit Control** will be called only when the *MU* is ready to exit the *Turntable*. This point in time is too late though, as the target has to be determined earlier in time.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

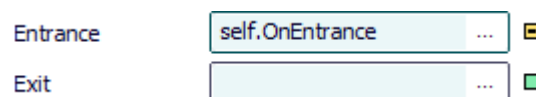
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.

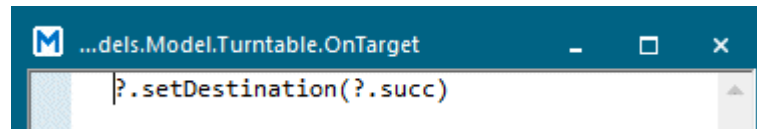


Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

The **standard target control** looks like this:



SimTalk

[TargetCtrl \[SimTalk\] - Turntable](#)

See also

[Select Object \[for controls\]](#)

[setDestination \[SimTalk\] - Turntable](#)

[getDestination \[SimTalk\] - Turntable](#)

Tab Exit

Select to which of its successors the object moves the *MU* on the [Tab Exit](#).

See also

[Blocking \[exit strategy\]](#)

[Strategy \[material flow objects\]](#)

Tab Statistics [Turntable]

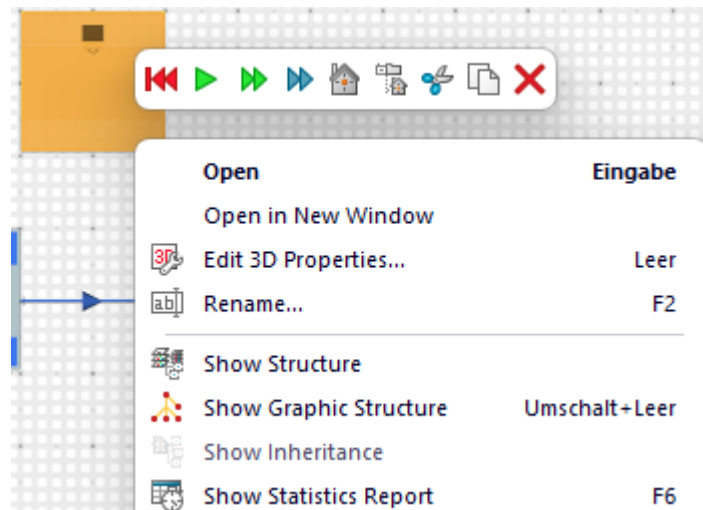
Statistics is described under the [Tab Statistics](#).

In addition, the *Turntable* collects these statistical values:

Item English	Description	Read-only attribute	Item German
Rotation Empty	Shows the portion of the statistics collection period during which the object was rotating Empty , i.e., without moving a part.	StatRotationEmptyPortion [SimTalk] - Turntable	Drehung leer
Rotation Loaded	Shows the portion of the statistics collection period during which the object was rotating while moving a part.	StatRotationLoadedPortion [SimTalk] - Turntable	Drehung belegt

Statistics					
Resource Type: Transport					
☒ Resource Statistics					
Working	100.00%	Rel. Occupation	14.34%	Contents	1
Setting-up	0.00%	Rel. Empty	19.31%	Min. Contents	0
Waiting	0.00%			Max. Contents	1
Blocked	0.00%	Entr. Blocked	94.88%	Entries	265106
P. up/down	0.00%	Exit Blocked	0.00%	Exits	265105
Failed	0.00%	Rotation Empty	14.19%		
Stopped	0.00%	Rotation Loaded	26.43%		
Paused	0.00%				
Unplanned	0.00%	Average Dwell Time	13.4232		

To view **Rotation Time** in the **Statistics Report**, select **View > Show Statistics Report** in the dialog of the object. You can also click the right mouse button in the *Frame* and select **Show Statistics Report** or you can press **F6**.



See also

[Resource Statistics \[check box\]](#)

[Resource Type](#)

Tab Energy

Select **energy settings** for the object on the [Tab Energy](#).

Tab Costs

Select **costs settings** for the object on the [Tab Costs](#).

While the *Turntable* transports *MUs* costs accrue which result from the sum of the total investment costs and the total operating costs.

Note

The total investment costs only accrue during the **Depreciation Period**.

If the *Turntable* is empty, the costs remain with the *Turntable* as **general costs**.

See also

[Simulate the Accrued Costs of the Machines](#)

[CostAnalyzer](#) > [How the CostAnalyzer Assigns Costs to Part Types](#)

[CostAnalyzer](#) > [Costs Shown in the Costs Report](#)

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

The **View Menu** provides commands to access its functions.

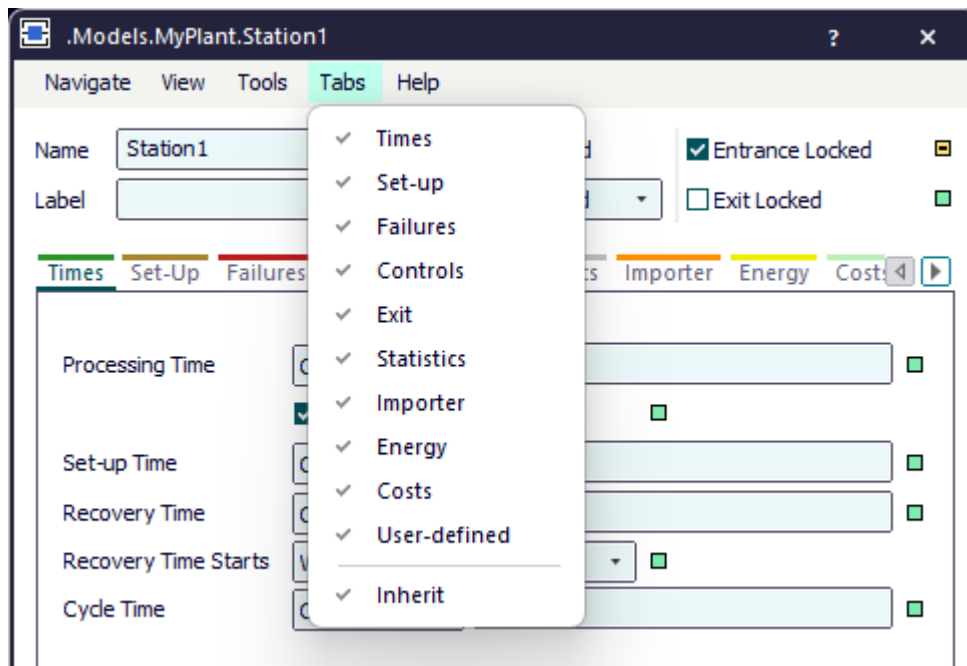
Refresh [on View menu]	Backward Blocking List
Show Statistics Report [on View menu]	Exit Blocking List
Show Attributes and Methods [on View menu]	Associated Lockout Zones
Contents [material flow objects]	Associated Shift Calendar
Forward Blocking List	

Tools Menu

The commands are described under the [Tools Menu](#).

Tab Menu

Use the commands of the **Tab** menu to show or hide individual tabs of the selected material flow objects. If you hide tabs that you do not need, *Plant Simulation* opens the dialog faster, and you can change to those tabs faster that you need in your daily work.



Remarks

To apply the changed settings, click **OK**, close the dialog, and reopen it

The menu shows a check mark to the left of the displayed tabs.

The command **Inherit** turns inheritance of the displayed or hidden tabs in the dialog off or on.

Help Menu

The commands are described under the [Help Menu](#).

Methods of the Turntable

_Methods of the Turntable

The *Turntable* provides:

- The *methods* listed in the table of contents to the left.
- The [_Methods of Curved Objects](#).
- The [Methods of the Material Flow Objects](#).
- The [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			([Contents:table]) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				([IconName:string, Width:integer, Height:integer]) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of
the method

Identifier and data type of
the parameter you enter

Data type of the
return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			([Index:integer:=1]) -> object
MUPart	VOID			([Index:integer:=1]) -> object

Name of
the method

Identifier and data type of
the parameter you enter

Default
value

Data type of the
return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).